SRI SARADI SCIENTIFIC PRIVATE LIMITED

Hyderabad, India

CUSTOM BIOLOGICS SERVICES CATALOGUE

Peptide Synthesis · Protein and Antibody Labeling
Recombinant Protein and Antibody Expression
Transmembrane Protein Platforms · Small-Molecule Labeling

For technical discussion and project quotation
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ABOUT THIS CATALOGUE

Sri Saradi Scientific supplies laboratory products and scientific services to research, development and quality control laboratories across India. Alongside our product portfolio we place custom synthesis, labeling and protein expression projects with an established manufacturing partner, giving Indian laboratories access to capability that is otherwise difficult to source locally.

This catalogue describes what we can undertake. Every project is scoped individually, because scale, purity, modification and delivery format all change what is required at the bench. Send us a sequence, a target, or a description of the material you need worked on, and we will return a written proposal with scope, deliverables, quality control data included, and a realistic delivery date.

Services at a glance

- **1.** Enzyme, biotin and fluorophore labeling of antibodies and large proteins
- **2.** Biotin, fluorophore and stable isotope labeling of small molecules, and custom small-molecule synthesis
- **3.** Custom peptide synthesis, from standard linear peptides to complex modified and cyclic sequences
- **4.** Peptide libraries for screening and epitope mapping
- **5.** Oligonucleotide synthesis with DSL, PAGE and HPLC purification
- **6.** Recombinant protein expression in prokaryotic systems, from gene synthesis to purified protein
- **7.** Recombinant antibody expression in CHO and HEK293 systems
- **8.** Multi-pass transmembrane protein expression using VLP, detergent and nanodisc platforms

01 ANTIBODY AND PROTEIN LABELING

Conjugation of your own antibody or protein with an enzyme, biotin or a fluorescent label, prepared to order and returned with an electronic labeling report. This service supports in-house ELISA development, western blotting, immunohistochemistry and multicolour flow cytometry panels where a commercially conjugated equivalent is not available.

Labels available

- **Enzyme:** Horseradish peroxidase (HRP), for direct and sandwich ELISA and western blot detection
- **Affinity tag:** Biotin, for streptavidin-based amplification and capture systems
- **Fluorophores:** FITC, RBITC, Cy series (Cy3, Cy5, Cy5.5, Cy7) and Alexa Fluor series

Accepted material

Antibody IgG, large proteins, small peptides and small-molecule compounds. Peptides and small molecules require a different chemistry because of their low molecular weight and are scoped case by case.

Sample requirements

- Lyophilised powder is preferred. Liquid samples should be at a concentration above 0.5 mg/ml.
- Samples containing sodium azide cannot be labeled with HRP.
- Carrier proteins and additives such as BSA or glycine in the sample may compromise the conjugation and should be discussed before submission.
- For large proteins, please supply molecular weight, isoelectric point, buffer composition and pH.
- Please declare any other special handling requirement in advance.

02 SMALL-MOLECULE LABELING AND CUSTOM SYNTHESIS

Labeling and custom preparation of small-molecule compounds, carried out by a team with a long track record in small-molecule bioactive compound research. Small molecules present a narrow window for derivatisation, so each request is assessed on structure before a route is proposed.

Scope of work

- Biotin labeling of small-molecule compounds, for pull-down and target identification work
- Fluorophore labeling of small-molecule compounds, for imaging, binding and localisation studies
- Stable isotope labeling of small-molecule compounds, for mass spectrometry and metabolic tracing
- Custom synthesis of small-molecule compounds, where a synthetic route or supporting literature is provided

What to send us

The compound structure, the position at which labeling is required or preferred, the intended downstream application, and the quantity and purity you need. For custom synthesis, a proposed route or the relevant literature reference is required before the project can be assessed.

03 CUSTOM PEPTIDE SYNTHESIS

Peptides are prepared on automated synthesisers by a team with close to a decade of peptide design and synthesis experience. Building blocks and resins are selected to suit the sequence, coupling conditions are optimised, and alternative strategies are held in reserve for difficult sequences. Every purified peptide is delivered with complete quality control data, including HPLC and mass spectrometry reports.

Typical applications include immunogen preparation for antibody development, drug discovery and screening, enzyme substrate and inhibitor studies, and cell function analysis.

Standard linear peptides

Sequences of 2 to 30 amino acids, synthesised efficiently and delivered quickly, with purity achievable above 99%.

Complex linear peptides

Long peptides, poorly soluble hydrophobic peptides, and sequences with internal repeats, undertaken with a success rate above 90%.

Simple modifications

N-terminal acetylation and C-terminal amidation are provided at no additional charge.

Complex modifications

- **Fluorescent labels:** FAM, FITC, Rhodamine B, TAMRA, Cy3, Cy5, Cy5.5, Cy7, AFC, AMC and pNA
- **Biotinylation:** Biotin and Lys(Biotin)
- **Fatty acid modification:** Palmitic acid (Pal), myristic acid (Myr) and stearic acid (Ste)
- **Phosphorylation:** pSer, pThr and pTyr
- **Other modifications:** Sulfation, methylation, PEGylation, cholesteryl modification and lysine side-chain modification
- **Cyclic peptides:** Disulfide cyclisation and amide cyclisation

Unusual amino acids and derivatives

D-amino acids, pyroglutamic acid, hydroxyproline (Hyp), ornithine (Orn), 2-aminoisobutyric acid (Aib), norleucine (Nle), beta-alanine, 6-aminohexanoic acid (Ahx) and sarcosine (Sar).

Carrier protein conjugation

Conjugation to KLH, BSA or OVA, for immunisation and for use as a coating antigen in screening assays.

Purity grades offered

Crude peptide, desalted, and purified to >70%, >75%, >80%, >85%, >90%, >95% or >98%.

04 PEPTIDE LIBRARIES

A peptide library is a defined collection of related peptides prepared and delivered as a set. Libraries are the standard route to high-throughput screening in biopharmaceutical research and proteomics, and are used in drug development, vaccine development, immunotherapy, GPCR ligand screening, protein-protein interaction studies, protein function analysis, nucleic acid binding studies, enzyme substrate and inhibitor screening, antigen epitope mapping and signalling studies.

Specification	Crude peptide library	Purified peptide library
Length	5 to 25 amino acids	5 to 25 amino acids
Purity	Crude lyophilised powder	70% to 98%
Quantity	1 mg or more per peptide	1 mg or more per peptide
Delivery format	Lyophilised powder in individually labelled amber vials	Lyophilised powder in individually labelled amber vials
Quality control	HPLC and MS reports provided	HPLC and MS reports provided
Support	Full pre-sales and post-sales technical support	Full pre-sales and post-sales technical support

Library design is part of the service. Send us the parent sequence, the overlap and step size you have in mind, or simply the question you are trying to answer, and we will propose a library layout before quoting.

05 OLIGONUCLEOTIDE SYNTHESIS

Custom primers and oligonucleotides synthesised at a range of scales, with three purification routes so that the specification matches the application rather than defaulting to the most expensive option.

Purification methods

DSL purification uses a simple reverse-phase cartridge on which DNA is specifically adsorbed. The oligonucleotide is eluted with organic solvent and is not removed by water, so salt is cleared effectively. It does not, however, remove failure sequences shorter than the target. Suitable for routine PCR and sequencing primers.

PAGE purification is an optimised denaturing polyacrylamide gel electrophoresis process that separates DNA fragments at high purity and recovers the target sequence. Suitable for long oligonucleotides, cloning, and applications where truncated products cannot be tolerated.

HPLC purification separates the oligonucleotide on the basis of hydrophobicity by high-performance liquid chromatography. An efficient route to very high purity, suitable for probes, modified oligonucleotides and demanding downstream chemistry.

Specification

- 11 to 90 bases with DSL purification; 11 to 90 bases with PAGE or HPLC purification
- Supplied at 2 OD, 5 OD and 10 OD scales
- Longer sequences, modifications and probe chemistries are quoted on enquiry

06 RECOMBINANT PROTEIN EXPRESSION

A complete route from sequence to purified protein, delivered with a full experimental report. The service is modular: you may commission the whole workflow or only the steps you need.

Workflow

- **Gene synthesis.** Genes are synthesised to your sequence for constructs up to 4500 bp
- **Vector construction and subcloning.** Cloning into an expression vector chosen for the target, including subcloning from your existing construct
- **Pilot expression.** Small-scale expression and verification, so that solubility and yield are established before committing to a full run
- **Scale-up expression and purification.** Production and purification of the target protein at working quantities

Tags

Protein is supplied with a HIS tag as standard. Tag removal is available as an additional step where the downstream application requires an untagged product.

What you supply

- The protein or gene sequence
- Template material for PCR amplification where applicable, such as plasmid, cells or tissue, free of biosafety concerns

What you receive

- The purified recombinant protein
- A written experimental report covering construction, expression and purification

Mammalian cell expression

Expression of recombinant proteins in mammalian cell systems is available and is quoted on enquiry.

07 RECOMBINANT ANTIBODY EXPRESSION

Transient expression of double-chain recombinant antibodies in CHO or HEK293 systems from sequences you provide. This is the practical route to material for characterisation, assay development and early evaluation without establishing a stable line.

Specification

- **Expression systems:** CHO or HEK293, selected to suit the construct
- **Culture scales:** From 10 ml through 30, 50, 100, 200, 500 and 1000 ml; volumes above 1 litre quoted on enquiry
- **Format:** Double-chain antibody, purified and supplied with an experimental report

Suitable for

- Antibody discovery and lead characterisation
- Assay and diagnostic kit development requiring defined antibody material
- Comparability and reference material for analytical method development

08 TRANSMEMBRANE PROTEIN EXPRESSION

Between 20% and 30% of the genes in the genome encode membrane proteins, and these proteins mediate almost every cellular process, including ligand-receptor binding, signal transduction, molecular transport, cell-cell recognition and enzymatic catalysis. Ion channels, transporters, GPCRs and kinases together form the largest single class of drug targets, and roughly 40% of approved antibody drugs act through GPCR complexes.

Full-length multi-pass membrane proteins are among the hardest targets to prepare in a usable form. Three complementary platforms are available so that the preparation can be matched to the assay rather than the assay compromised to fit the preparation.

Criterion	VLP	Detergent	Nanodisc
ELISA coating	Direct coating	Direct coating	Direct coating
Activity	Highest	Good	Good
Stability	Highest	Good	Good
Purity	Mixture	Target protein only	Mixture

Criterion	VLP	Detergent	Nanodisc
SPR and BLI	Compatible, with conditions	Compatible, solution requires adjustment	Compatible
Points to note	Stated concentration is total protein; higher overall cost	Shorter preparation time, more economical	Shorter preparation time, more economical

Platform strengths

- **Expression strategy.** Three expression platforms paired with optimised vectors for efficient expression of multi-pass membrane proteins
- **Experience.** More than ten years of recombinant protein expression and over 10,000 recombinant proteins expressed
- **Quality.** Strict quality control, delivering protein of high purity and retained activity suitable for downstream study
- **Yield.** Up to 5 mg/ml in the cell-free system and up to 100 mg/L in the mammalian cell system

HOW WE WORK

1. Enquiry and scoping

Contact us with the sequence, target or material and the application you have in mind. A technical discussion establishes what is feasible, what specification is appropriate, and what the project will involve. For routine peptide enquiries a full quotation is normally returned within one working day.

2. Proposal and agreement

You receive a written proposal covering scope, deliverables, quality control data included, and delivery date. The project proceeds on a signed agreement between us.

3. Sample submission

Where the project requires your own material, the sample is submitted with a completed sample receipt form. We will advise on packaging, cold chain and documentation before anything is dispatched.

4. Execution and reporting

Work proceeds to the agreed specification. For larger projects a project manager reports progress in writing or by call at weekly or fortnightly intervals, and we remain in contact through the design stage, at key execution points, and for a week after delivery.

5. Delivery

Material is delivered as lyophilised powder or in the agreed format, with the full quality control package. Purified peptides are supplied with HPLC and mass spectrometry reports; labeling projects are supplied with a labeling report; expression projects are supplied with a written experimental report.

6. Technical support

Pre-sales and post-sales technical support is part of the service. If the material does not perform as expected in your hands, we would rather hear about it and work through it than leave the question open.

CONFIDENTIALITY

Customer material and customer information are treated as confidential. A non-disclosure agreement can be executed before any sequence, structure or sample is disclosed, and we are willing to be a named party to that agreement rather than stand outside it. Sequences, structures and project details are not disclosed to any third party and are not used for any purpose other than the commissioned work.

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All services are quoted on enquiry. Prices are available on request.